

MKTECH: Offering glitchless media streaming experiences to global users

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ABOUT MKTECH

Leveraging Google
to deploy its digital
services, MKTECH
improve its bandwidth
quality and reduce
operational workload
enhancing product
development efficiency

MKTECH

About MKTECH

Founded in 2004 and headquartered in Zhuhai, China, MKTECH is a home smart device hardware and software provider specializing in digital media, smart IoT and smart manufacturing. Producing 30 million digital end devices every year and offering one-stop software solutions, MKTECH is dedicated to bringing intelligent and safe life to every household around the world. It possesses more than 100 intellectual property and patents.

Industries: Media & Entertainment

Google Cloud results

Helps reduce the number of user reports by improving bandwidth quality through GKE and Cloud Load Balancing

Increases product development efficiency by 20% with cloud-native tools of GKE

Enhances video transcoding and decoding efficiency by 4-5x with Cloud GPUs

Founded in 2004, MKTECH is a hardware and software provider with end devices

Reduce operational workload

Location: China

CDN (<https://cloud.google.com/cdn/>), Media



About Baidao

Baidao is a cloud management solution provider and a Google Cloud Premier Partner headquartered in Hong Kong with great experiences in public cloud management. Offering operation and consulting services for cloud migration, big data & AI platforms and cloud security, Baidao has more than 200 enterprise customers across various industries.



Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

Media CDN (<https://cloud.google.com/media-cdn>)

Cloud GPUs (<https://cloud.google.com/gpu/>)

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Looker (<https://cloud.google.com/looker>)

like set-top boxes, routers and fusion gateways as its flagship products. Specializing in digital media, smart IoT and smart manufacturing, MKTECH also offers one-stop solutions for smart device management.

MKTECH's main business includes providing custom digital TV and Internet of Things (IoT) solutions. Its Internet Protocol television (IPTV) and over-the-top (OTT) platforms not only enable regular media live streaming, but also support additional features like video on demand (VOD), gaming, ecommerce and karaoke. Besides, MKTECH's IoT platform can integrate multiple smart home devices to meet users' diverse needs. For example, by connecting a set-top box with a smart camera and a smart door lock, MKTECH's end users can check the security of their homes on smartphones.

After years of business expansion, MKTECH currently sees 80% of their sales coming from outside China, with most of its customers in Europe, Latin America and Southeast Asia. The company used to rent on-prem servers around the world to provide digital TV and IoT services to local users. However, as digital TV users started to watch more VOD programs than broadcast, MKTECH's digital TV system required higher bandwidth quality, which made it challenging for the company to ensure glitchless media streaming experiences globally.

To improve the user experience of its digital TV system, MKTECH in 2021 decided to migrate its services to a global public cloud platform.

Considering that [Google Cloud](#)

(<https://cloud.google.com/>) provides cloud infrastructure worldwide and can be seamlessly integrated with Google Maps Platform (<https://mapsplatform.google.com/>) and Google Home (<https://home.google.com/welcome/>), which MKTECH planned to leverage in the future to enrich its services, the company eventually chose to move its VOD, live streaming and IoT platforms to Google Cloud.

"Since the users of our digital TV and IoT services are all over the world, we need powerful global infrastructure to enable quality user experiences," says Rongke Pan, Product Director at MKTECH. "Not only does Google Cloud meet this need, its high compatibility with other products of Google also enhances our ability to expand business. For example, Google Maps Platform can send back location data of our IoT devices, while Google Home can be integrated with our smart home products."

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*—Rongke Pan, Product
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Enhancing development efficiency with an optimized DevOps structure in the cloud

Before the migration, Google Cloud partner [Baidao](https://www.baidadata.com/) (<https://www.baidadata.com/>) provided detailed explanations regarding cost estimation and architecture optimization plans to MKTECH. With the technical support of the Google Cloud and Baidao team, MKTECH moved its services to Google Cloud within six months and changed its DevOps structure to improve its development efficiency.

Pan points out that due to the unclear structure, MKTECH's previous DevOps process faced many challenges, including lagging progress, inefficient cross-team communication and inadequate use of DevOps products or tools. After the migration to Google Cloud, the Baidao team has helped MKTECH establish a training mechanism for the whole product development cycle and set up a more streamlined, structured DevOps process in Google Cloud.

"In the past, we didn't have good infrastructure planning. Google Cloud provides many excellent tools enabling us to optimize our infrastructure architecture and build a more efficient DevOps workflow. During the process, Baidao's technical advice has given us great help," adds Pan.

Improving bandwidth quality while reducing operational costs with GKE

Now, MKTECH uses [Google Kubernetes Engine](https://cloud.google.com/gke) (<https://cloud.google.com/gke>) (GKE) to deploy its containerized applications. Pan says that as a fully managed Kubernetes tool, GKE provides a highly scalable, flexible, secure and stable operating environment that has effectively enhanced MKTECH's service quality and operational efficiency.

Before the migration, MKTECH's VOD services deployed in on-prem servers were frequently inaccessible during high traffic times. The company's operations team then needed to manually add new computing resources, which was

time- and effort-consuming. With the Autopilot

(<https://cloud.google.com/kubernetes-engine/docs/concepts/autopilot-overview>)

mode of GKE, the containers that support MKTECH's VOD services in GKE would automatically scale up within a few seconds after the bandwidth reaches 1G. As a result, the MKTECH team's workloads of managing and maintaining servers have reduced by 80%. Besides, MKTECH also leverages Cloud Load Balancing (<https://cloud.google.com/load-balancing>) to further improve its bandwidth quality.

"The autoscaling feature of GKE enables us to manage traffic peaks for our VOD services with minimal labor costs, ensuring stable and glitchless media streaming experiences. Before the migration, our operational team needed to be on call around the clock. Now, no one has to do night shifts anymore," notes Pan.

On top of that, the development tools and APIs of GKE have also helped MKTECH further enhance its development efficiency. While developing the "system list" module, the MKTECH team noticed that using cloud-native tools of GKE saved 20% of development time. This means that the company is able to complete development projects faster, and its engineers have more time to invest in developing innovative products.

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Enabling smooth media streaming experiences around the world with Cloud CDN and Media CDN

Since 80% of MKTECH's business is outside China, it needs global content delivery tools to offer quality media streaming services to users around the world.

The company currently leverages Cloud CDN and Media CDN (<https://cloud.google.com/cdn>) to support its media streaming services. With the globally distributed network nodes of Google Cloud, MKTECH is able to reach ideal data transferring speed and response time for global users.

"Cloud CDN and Media CDN have respective advantages. Cloud CDN delivers excellent performance for access to software updates and static images, while Media CDN empowers speedy delivery of video files. Using both tools enables us to provide smooth media streaming services," says Pan.

Furthermore, the computing resources of Google Cloud also allows MKTECH to continue upgrading its media streaming services. With the continuous upgrades of camera and display devices, 4K and 8K videos have become mainstream in the streaming media industry. To transcode and decode videos with higher resolution, more efficient computing resources are required. In this regard, Cloud GPUs have enhanced MKTECH's video transcoding and decoding efficiency by 4-5x.

Optimizing smart home experiences with AI and ML

Next, MKTECH will employ the data analytics, artificial intelligence (AI) and machine learning (ML) capabilities of Google Cloud to further optimize its services. The team is testing using [BigQuery](https://cloud.google.com/bigquery) (<https://cloud.google.com/bigquery>) and [Looker](https://cloud.google.com/looker) (<https://cloud.google.com/looker>) to analyze the user behavior data of its media streaming services to generate personalized content recommendations for a higher free to paid conversion rate. The Google ecosystem has facilitated the development of the project, with Media CDN being seamlessly integrated with the data tools, and Google Maps Platform sending location data instantly to BigQuery, which allows the MKTECH team to understand the popularity of each show in different regions.

Besides, MKTECH will use the latest AI and ML technologies of Google Cloud to develop interactive home robots. Pan notes that the industry-leading capabilities of Google Cloud in big data and large language models support more natural conversations between humans and robots, leading to better IoT user experiences.

He adds, "Google Cloud has helped us optimize our DevOps structure and effectively improve the quality of our media streaming and IoT services outside China. In the future, we'll take advantage of the AI capabilities of Google Cloud to advance our smart home technology, enhance user experiences, and create greater values for our customers."

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